

1

BASIC CALCULUS

1.1 Introduction

1.1.1 Limits, Continuity and Differentiability

a. As x tends to a ($x \rightarrow a$) $\Rightarrow x$ is moving towards a

A value l is said to be limit of a function $f(x)$ at $x \rightarrow a$ if $f(x) \rightarrow l$ as $x \rightarrow a$.

It is mathematically defined as

$$\lim_{x \rightarrow a} f(x) = l = \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

A function $f(x)$ is said to be continuous at $x = a$ if

$$\lim_{x \rightarrow a} f(x) = l = f(a) = f(x)|_{x=a}$$

Note:

For $\lim_{x \rightarrow a} f(x)$ to exist, the function need not be continuous at $x = a$.

But for $f(x)$ to be continuous at $x = a$, $\lim_{x \rightarrow a} f(x)$ should exist.

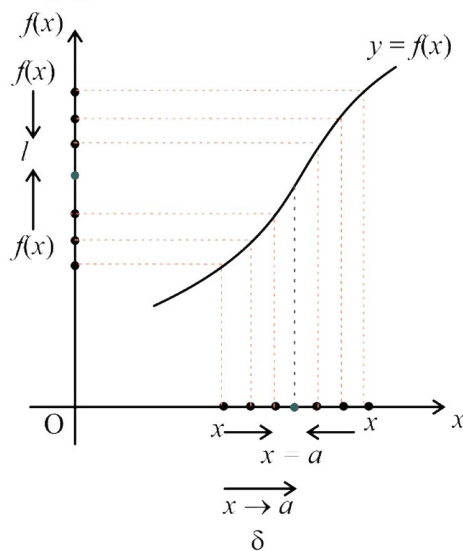


Fig. 1.1

b. Concept of differentiability

A continuous function $f(x)$ is said to be differentiable at $x = a$ if $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists.

$$f'(x)|_{x=a} = f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$f'(a) = \tan \theta$ where θ is the angle made by the tangent to the curve at $x=a$ with x – axis.

c. Some Standard Derivatives

$$(i) \quad \frac{d}{dx}(x^n) = n \cdot x^{n-1}$$

$$(ii) \quad \frac{d}{dx}(\sin x) = \cos x$$

$$(iii) \quad \frac{d}{dx}(\cos x) = -\sin x$$

$$(iv) \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

$$(v) \quad \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$(vi) \quad \frac{d}{dx}(\sec x) = \sec x \cdot \tan x$$

$$(vii) \quad \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$(viii) \quad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}; -1 < x < 1$$

$$(ix) \quad \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, -1 < x < 1$$

$$(x) \quad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$(xi) \quad \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$(xii) \quad \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$$

$$(xiii) \quad \frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}; |x| > 1$$

$$(xiv) \quad \frac{d}{dx}(\log_a x) = \frac{1}{x \log_e a}$$

$$(xv) \quad \frac{d}{dx}(\log_e x) = \frac{1}{x}$$

$$(xvi) \quad \frac{d}{dx}(a^x) = a^x \cdot \log_e a$$

$$(xvii) \quad \frac{d}{dx}(e^x) = e^x$$

$$(xviii) \quad \frac{d}{dx}(|x|) = \frac{|x|}{x} (x \neq 0)$$

$$(xix) \quad \frac{d}{dx}(x^x) = x^x(1 + \log_e x)$$

$$(xx) \quad \frac{d}{dx}(\sinh x) = \cosh x$$

d. Product rule of differentiation

$$\frac{d}{dx}(f(x).g(x)) = f(x).g'(x) + f'(x).g(x)$$

$$d(uvw) = uvw' + uv'w + u'vw$$

e. Quotient rule of differentiation

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x).f'(x) - f(x).g'(x)}{(g(x))^2}, (g(x) \neq 0)$$

f. Greatest Integer function / step function / integer part function

$$f(x) = [x] = n, \forall n \leq x < n + 1 \text{ where } n \in \mathbb{Z}$$

$$\lim_{x \rightarrow a} [x] = \nexists \text{ if } a \text{ is an integer}$$

$$\text{L.H.L.} = \lim_{x \rightarrow a^-} [x] = a - 1$$

$$\text{R.H.L.} = \lim_{x \rightarrow a^+} [x] = a$$

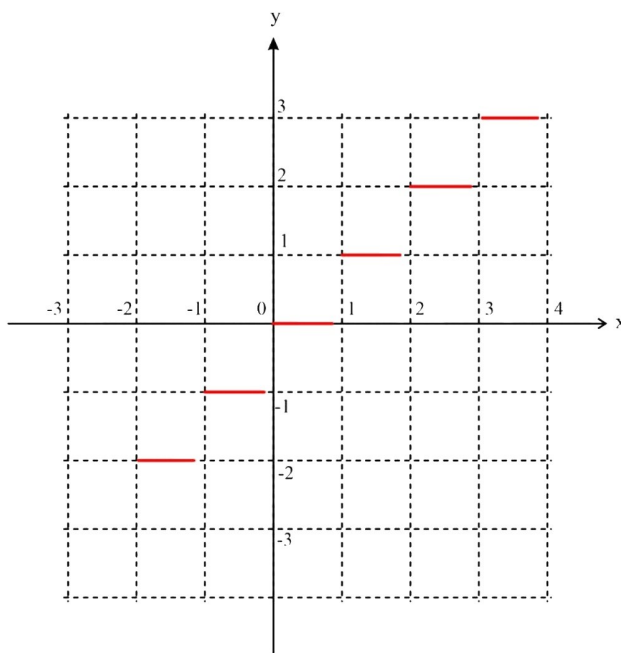


Fig.1. 2 Greatest Integer

g. Properties of Limits

$$(i) \quad \lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

$$(ii) \quad \lim_{x \rightarrow a} (f(x).g(x)) = \lim_{x \rightarrow a} f(x). \lim_{x \rightarrow a} g(x)$$

$$(iii) \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \left(\lim_{x \rightarrow a} g(x) \neq 0 \right)$$

(iv) If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} g(x) = \nexists$, then $\lim_{x \rightarrow a} f(x) \cdot g(x)$ MAY exist

Ex: let $f(x) = \sin x$, $g(x) = \frac{1}{x}$, $\lim_{x \rightarrow 0} f(x) = 0$, $\lim_{x \rightarrow 0} \frac{1}{x} = \nexists$

But $\lim_{x \rightarrow 0} \sin x \cdot \frac{1}{x} = 1$

(v) If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ (or) $\frac{\infty}{\infty}$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \neq \left(\frac{0}{0}\right)$

If $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \frac{0}{0}$ (or) $\frac{\infty}{\infty}$, then $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)}$ and so on

(vi) If $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = 0 \times \infty \Rightarrow \lim_{x \rightarrow a} \frac{f(x)}{\left(\frac{1}{g(x)}\right)} = \frac{0}{0}$ (Apply L- Hospital Rule again)

h. Some Standard Limits

(i) $\lim_{x \rightarrow a} \frac{\sin x}{x} = 1$

(ii) $\lim_{x \rightarrow a} \frac{\tan x}{x} = 1$

(iii) $\lim_{x \rightarrow 0} \frac{1 - \cos ax}{x^2} = \frac{a^2}{2}$

(iv) $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$

(v) $\lim_{x \rightarrow \infty} \frac{\cos x}{x} = 0$

(vi) $\lim_{x \rightarrow 0} (1 + ax)^{b/x} = e^{ab}$

(vii) $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^{bx} = e^{ab}$

(viii) $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2}\right)^{1/x} = \sqrt{ab}$

(ix) $\lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x + \dots + n^x}{n}\right)^{1/x} = \sqrt[n]{n!}$

(x) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$; $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$

(xi) $\lim_{x \rightarrow 0} x \cdot \sin\left(\frac{1}{x}\right) = 0$

1.2 Mean Value Theorems

1.2.1 Lagrange's Mean Value Theorem (LMVT):

If $f(x)$ is continuous in $[a, b]$ and it is differentiable in (a, b) then $\exists$ at least one point 'C' such that $C \in (a, b)$ and

$$f'(C) = \frac{f(b) - f(a)}{b - a}$$

Here $f'(C)$ slope of tangent to $f(x)$ at $x = C$.

Tangent at $x = c$ is parallel to the line connecting the points A and B

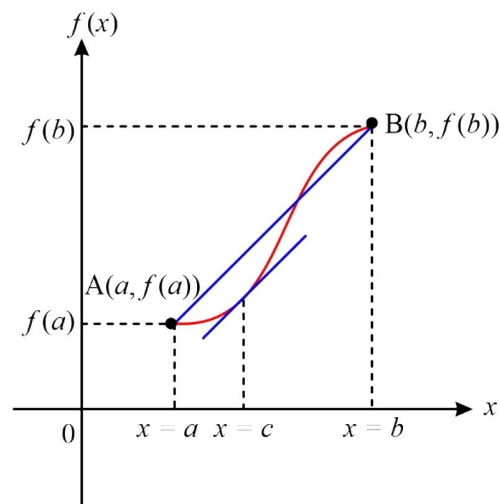


Fig.1.3 LMVT

1.2.2 Rolle's Mean Value Theorem

If $f(x)$ is continuous in $[a, b]$ and differentiable in (a, b) and $f(a) = f(b)$ then $\exists$ at least one-point $C \in (a, b)$ such that $f'(C) = 0$.

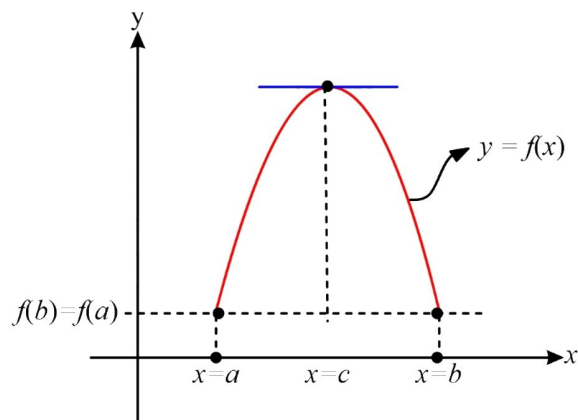


Fig.1.4 Rolle's mean value

1.2.3 Cauchy's Mean Value Theorem

If $f(x)$ and $g(x)$ are continuous in $[a, b]$ and differentiable in (a, b) then $\exists$ at least one value of 'C' such that $C \in (a, b)$ and $\frac{g'(C)}{f'(C)} = \frac{g(b)-g(a)}{f(b)-f(a)}$

1.3 Increasing and Decreasing Functions

1.3.1 Increasing Functions

A function $f(x)$ is said to be increasing if $f(x_1) < f(x_2) \forall x_1 < x_2$

Or

A function $f(x)$ is said to be increasing if $f(x)$ increases as x increases.

For a function $f(x)$ to be increasing at the point $x=a$, $f'(a) > 0$.

Example:

$e^x, \log_e x \rightarrow$ Monotonically increasing functions

$\sin x$ in $(0, \pi/2) \rightarrow$ non-monotonic functions

1.3.2 Decreasing Functions

A function $f(x)$ is said to be a decreasing function if $f(x_1) > f(x_2) \forall x_1 < x_2$

A function $f(x)$ is said to be decreasing function if $f(x)$ decreases as x increases.

Example: $e^{-x} \rightarrow$ Monotonically decreasing function $\sin x$ in $(\frac{\pi}{2}, \pi)$

1.4 Concept of Maxima and Minima

Let $f(x)$ be a differentiable function, then to find the maximum (or) minimum of $f(x)$.

(1) Find stationary points from the equation $f'(x) = 0$. Let ' x_0 ' be the stationary point.

(2) Find the value of $f''(x_0)$

Case (i): If $f''(x_0) < 0$, then the function $f(x)$ has maximum at $x = x_0$ and the maximum value of the function is $f(x_0)$.

Case (ii): If $f''(x_0) > 0$, then the function $f(x)$ has minimum value at $x = x_0$ and the minimum value is $f(x_0)$.

Case (iii): If $f''(x_0) = 0$, then we cannot comment on the existence of maximum (or) minimum of $f(x)$ at $x = x_0$.

Such points are called points of inflection (or) Critical points.

Example: $x = 0$ is a critical point of $f(x) = x^3$

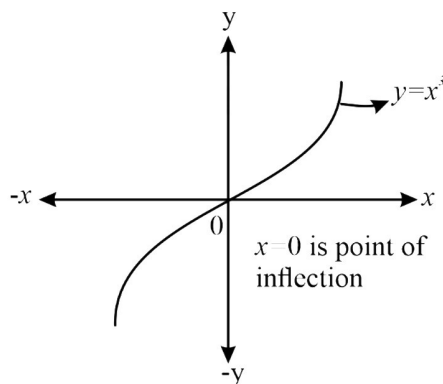


Fig. 1.5 Graph of x^3

$$f(x) = x^3$$

$$f'(x) = 3x^2 = 0 \Rightarrow x = 0$$

$$f''(x) = 6x \Rightarrow f''(0) = 6(0) = 0$$

$\Rightarrow$

1.5 Taylor Series

If $f(x)$ is continuously differentiable ($f'(x), f''(x), f'''(x), \dots$ exists) then the Taylor series expansion of $f(x)$ about the point $x = a$ is given by

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots \infty$$

$$\text{If } a = 0, \text{ then } f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots \infty$$

The coefficient of $(x-a)^n$ in the Taylor series expansion of $f(x)$ is $\frac{f^n(a)}{n!}$.

The general expansion of Taylor series is given by $f(x+h) = f(x) + h \cdot \frac{f'(x)}{1!} + h^2 \cdot \frac{f''(x)}{2!} + h^3 \cdot \frac{f'''(x)}{3!} + \dots \infty$

- Finding the expansion of e^x about $x = 0$

$$f(x) = e^x \Rightarrow f(0) = e^0 = 1$$

$$f'(x) = e^x \Rightarrow f'(0) = e^0 = 1; f''(0) = f'''(0) = f^{(4)}(0) = \dots = 1$$

$$f(x) = e^x = 1 + (x-0) \cdot \frac{1}{1!} + (x-0)^2 \cdot \frac{1}{2!} + (x-0)^3 \cdot \frac{1}{3!} + \dots$$

$\Rightarrow$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

1.6 Integral Calculus

If $F(x)$ is anti-derivative of $f(x)$ that is continuous and differentiable in (a, b) , then we write $\int_{x=a}^{x=b} f(x) dx = F(b) - F(a)$. Here $f(x)$ is integrand

If $f(x) > 0 \forall a \leq x \leq b$, the $\int_a^b f(x) dx$ represents the shaded area in the given figure.

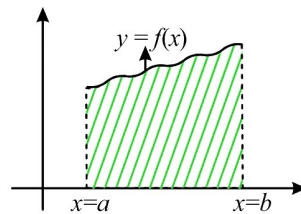


Fig.1.6 Integration of continuous function

1.6.1 Mean Value Theorem of Integration

If $f(x)$ is continuous in $[a, b]$ and differentiable in (a, b) then $\exists$ atleast one-point $C \in (a, b)$ such that

$$f'(C) = \frac{\int_a^b f(x) dx}{(b-a)}$$

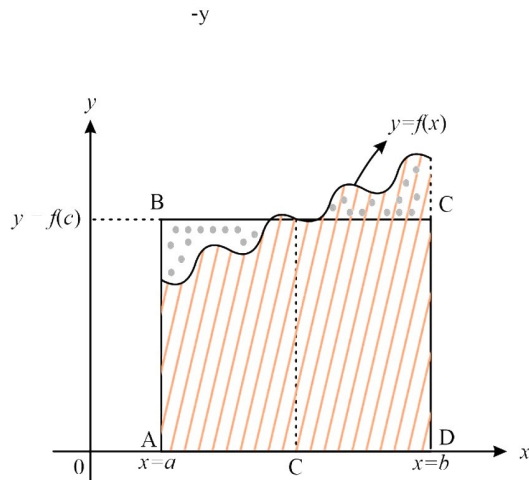


Fig.1.7 Mean value of integration

1.7 Newton-Leibnitz Rule

If $f(x)$ is continuously differentiable and $\phi(x)$, $\Psi(x)$ are two functions for which the 1st derivative exists, then

$$\frac{d}{dx} \left(\int_{\phi(x)}^{\psi(x)} f(x) dx \right) = f(\psi(x)) \cdot \psi'(x) - f(\phi(x)) \cdot \phi'(x)$$

Ex. $\frac{d}{dx} \left(\int_x^{x^2} \sin x \, dx \right) = \sin(x^2) \cdot 2x - \sin x \cdot 1 = 2x \sin(x^2) - \sin x$

1.8 Some Standard Integrals

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C (n \neq -1)$
2. $\int \frac{1}{x} dx = \log_e |x| + C$
3. $\int \sin x \, dx = -\cos x + C$
4. $\int \cos x \, dx = \sin x + C$
5. $\int \frac{f'(x)}{f(x)} dx = \log_e |f(x)| + C$
6. $\int \tan x \, dx = -\int -\frac{\sin x}{\cos x} dx = -\log_e |\cos x| + C$
 $\Rightarrow \int \tan x \, dx = \log_e |\sec x| + C$
7. $\int \cot x \, dx = \int \frac{\cos x}{\sin x} dx = \log_e |\sin x| + C = -\log_e |\cos x| + C$
8. $\int \sec x \, dx = \log_e |\sec x + \tan x| + C$
 $\int \sec x \, dx = \int \frac{\sec x (\sec x + \tan x)}{(\sec x + \tan x)} dx = \log_e |\sec x + \tan x| + C$
9. $\int \csc x \, dx = \log_e |\csc x - \cot x| + C$
10. $\int a^x dx = \frac{a^x}{\log_e a} + C$

11. $\int \frac{1}{x \log_e a} dx = \log_a x + C$
12. $\int x^x (1 + \log_e x) dx = x^x + C$
13. $\int f(x) \cdot f'(x) dx = \frac{1}{2} (f(x))^2 + C$
14. $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2 \sqrt{f(x)} + C$
15. If $f(x), g(x)$ are two functions, that are differentiable, then

$$\int f(x) \cdot g(x) dx = f(x) \cdot \int g(x) - \int f'(x) (g(x) dx) dx + C$$

Before integrating the product, the functions $f(x)$ and $g(x)$ are to be arranged according to the ILATE Principle.

Here, ILATE stands for INVERSE LOGARITHMIC ALGEBRAIC TRIGONOMETRIC EXPONENTIAL.

1.9 Properties of Definite Integrals

1. If $f(x)$ is differentiable in interval (a, b) , then $\int_a^b f(x) dx = - \int_b^a f(x) dx$
2. If $\exists$ a point $C \in (a, b)$ such that $f(x)$ is not differentiable, then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$
3. If $f(x)$ is continuously differentiable function,

$$\int_{-a}^a f(x) dx = 2 \times \int_0^a f(x) dx; \text{ if } f(-x) = f(x)$$

$$\Rightarrow f(x) \text{ is even function"}$$

$$0; \text{ if } f(-x) = -f(x) (\Rightarrow f(x) \text{ is odd function})$$
4. $\int_0^{2a} f(x) dx = 2 \times \int_0^a f(x) dx$ if $f(2a - x) = f(x)$
5. $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$
6. $\int_a^b \frac{f(x)}{f(x) + f(a+b-x)} = \left(\frac{b-a}{2} \right)$

Ex.

- (i) $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} = \frac{\pi}{4}$
- (ii) $\int_0^{\pi/2} \frac{1}{1 + \sqrt{\tan x}} dx = \int_0^{\pi/2} \frac{1}{1 + \left(\frac{\sqrt{\sin x}}{\sqrt{\cos x}} \right)} dx = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx = \frac{\pi}{4}$
- (iii) $\int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} = \left(\frac{3-2}{2} \right) = \frac{1}{2}$
- (iv) $\int_0^{\pi/2} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx = \frac{\pi}{4}$
7. $\int_0^{\pi/2} \sin^m x dx = \int_0^{\pi/2} \cos^m x dx = \frac{(m-1) \times (m-3) \times (m-5)}{m \times (m-2) \times (m-4)} \times \dots \left(\frac{1}{2} \right) \text{ (or) } \frac{2}{3} \times K$

Where $K = \pi/2$ if m is even

$= 1$ if m is odd.

8. $\int_0^{\pi} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} = \frac{\pi}{ab}$
9. $\int_0^{\pi/2} \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} = \frac{\pi}{2ab}$

1.10 Length of a Curve

If $y = f(x)$ is a differentiable function in (a, b) , then the length L of the curve $y = f(x)$ between $x = a$ and $x = b$ is given by

$$L = \int_{x=a}^{x=b} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

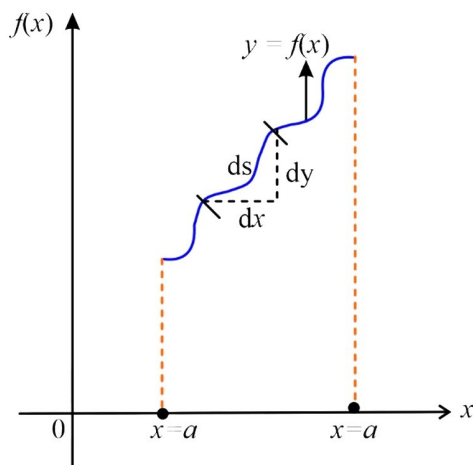


Fig.1.8 Length of the curve

1.11 Surface Area of Solid generated by revolving a curve about a fixed axis.

Elemental Surface Area

$$dA = 2\pi y \times ds = 2\pi y ds$$

$$\Rightarrow \text{Total surface area} = A = \int_{x=a}^{x=b} 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

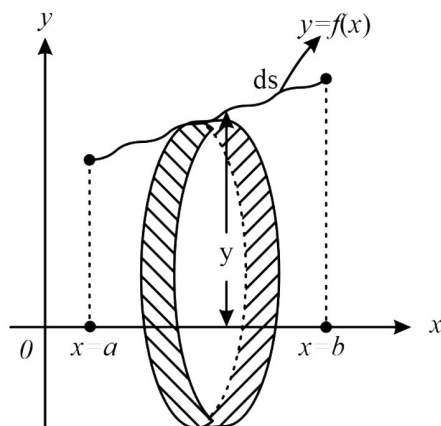


Fig.1.9 Surface area

1.12 Volume of the solid

The volume of the solid obtained by revolving the curve $y = f(x)$ between the lines $x = a$ and $x = b$ is given by

$$V = \int_{x=a}^{x=b} dV = \int_{x=a}^{x=b} \pi y^2 ds = \int_{x=a}^{x=b} \pi y^2 \cdot \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$\Rightarrow$

$$V \approx \int_{x=a}^{x=b} \pi y^2 dx$$

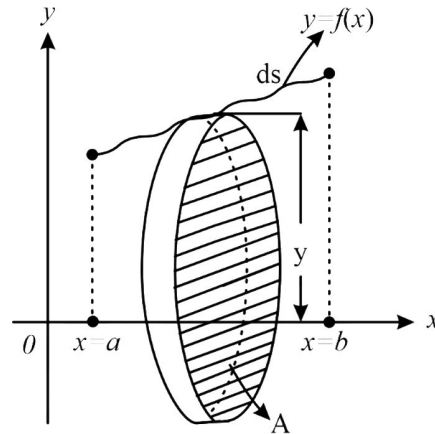


Fig.1.10 Volume of the solid

1.13 Gamma Function

The integral $\int_0^\infty e^{-x} \cdot x^{n-1} dx (n > 0)$ is called Gamma function of n . It is denoted by $\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$.

1.13.1 Properties of Gamma Function

- (i) $\Gamma n = (n - 1)!$
- (ii) $\Gamma(n + 1) = (n)!$
- (iii) $\Gamma(n + 1) = n\Gamma n$
- (iv) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

1.14 Beta Function

The function $\beta(m, n) = \int_0^1 x^{m-1} \cdot (1 - x)^{n-1} dx (m, n > 0)$ is called β function of m and n .

1.14.1 Properties of β function

- (i) $\beta(m, n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma(m+n)}$
- (ii) $\beta(m, n) = \beta(n, m)$
- (iii) $\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$
 $\beta(n, m) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx$

1.15 Area under the curves

If the function $f(x) > g(x)$ for all values of x between $x=a$ and $x=b$ then

$$A = \int_a^b f(x)dx - \int_a^b g(x)dx$$

⇒

$$A = \int_a^b (f(x) - g(x)) dx$$

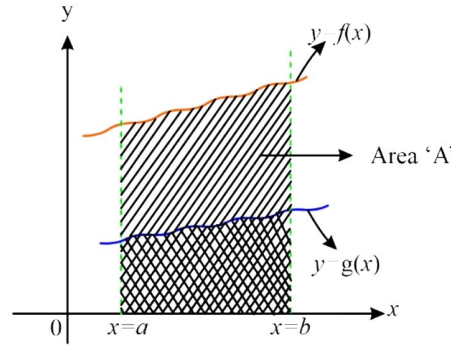


Fig.1.11 Area under curve

1.16 Multi Variable Calculus

a. Continuity of a function

A function $f(x, y)$ is said to be continuous at (a, b) if $\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) = f(a, b)$

b. Differentiation of a two-variable function

If $f(x, y)$ is a continuous function, then the derivative of $f(x, y)$ with respect to x treating y as constant is given by

$$= \frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

The derivative of $f(x, y)$ with respect to y treating x as constant is given by $q = \frac{\partial f}{\partial y} = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k}$

c. Homogenous Function

A function $f(x, y)$ is said to be homogenous function of degree 'n' if $f(kx, ky) = k^n \cdot f(x, y)$.

$$\text{Ex. } f(x, y) = x^3 - 3x^2y + 3xy^2 + y^3$$

$$\Rightarrow f(kx, ky) = (kx)^3 - 3(kx)^2(ky) + 3(kx) \cdot (ky)^2 + (ky)^3$$

$$= k^3(x^3 - 3x^2y + 3xy^2 + y^3)$$

$$= k^3 \cdot f(x, y) \Rightarrow f(x, y) \text{ is a homogenous function of degree '3'.$$

d. Euler's Theorem

If $f(x, y)$ is a homogeneous function of degree 'n' then

$$(i) \quad x \cdot \frac{\partial f}{\partial x} + y \cdot \frac{\partial f}{\partial y} = nf$$

$$(ii) \quad x^2 \cdot \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1)f$$

If $f(x, y) = g(x, y) + h(x, y) + \phi(x, y)$ where $g(x, y)$, $h(x, y)$ and $\phi(x, y)$ are homogenous functions of degrees m , n and p respectively, then

$$x \cdot \frac{\partial f}{\partial x} + y \cdot \frac{\partial f}{\partial y} = m \cdot g(x, y) + n \cdot h(x, y) + p \cdot \phi(x, y)$$

$$x^2 \cdot \frac{\partial^2 f}{\partial x^2} + 2xy \cdot \frac{\partial^2 f}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 f}{\partial y^2} = m(m-1) \cdot g(x, y) + n(n-1) \cdot h(x, y) + p(p-1) \cdot \phi(x, y)$$

e. Concept of Maxima and Minima in Two Variables

If $f(x, y)$ is a two-variable differentiable function, then to find the maxima (or) minima.

Step-1: Calculate $p = \frac{\partial f}{\partial x}$ and $q = \frac{\partial f}{\partial y}$ and equate $p = 0$, $q = 0$

Let (x_0, y_0) be a stationary point.

Step-2: Calculate r, s, t where $r = \frac{\partial^2 f}{\partial x^2} \Big|_{(x_0, y_0)}$; $s = \frac{\partial^2 f}{\partial x \partial y} \Big|_{(x_0, y_0)}$; $t = \frac{\partial^2 f}{\partial y^2} \Big|_{(x_0, y_0)}$

Case (i): If $rt - s^2 > 0$ and $r > 0$, then the function $f(x, y)$ has minimum at (x_0, y_0) and the minimum value is $f(x_0, y_0)$.

Case (ii): If $rt - s^2 > 0$ and $r < 0$, then the function $f(x, y)$ has maximum at (x_0, y_0) and the maximum value is $f(x_0, y_0)$.

Case (iii): If $rt - s^2 \leq 0$; then we cannot comment on the existence of maxima and minima.

Such stationary points where $rt - s^2 \leq 0$ are called **saddle points**.

f. Concept of Constraint Maxima and Minima (Method of Lagrange's unidentified multipliers).

If $f(x, y, z)$ is a continuous and differentiable function, such that the variables x, y and z are related/constrained by the equation $\phi(x, y, z) = C$ then to calculate the extreme value of $f(x, y, z)$ using Lagrange's Method of unidentified multipliers.

Step-1: Form the function $F(x, y, z) = f(x, y, z) + \lambda \{\phi(x, y, z) - C\}$ where λ is a multiplier.

Step-2: Calculate $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$ and $\frac{\partial F}{\partial z}$ and equate them to zero

Step-3: Equate the values of λ from the above 3 equations and obtain the relation between the variables x, y and z .

Step-4: Substitute the relation between x, y and z in $\phi(x, y, z) = C$ and get the values of x, y, z . Let them be (x_0, y_0, z_0) .

Step-5: Calculate $f(x_0, y_0, z_0)$

The value $f(x_0, y_0, z_0)$ is the extreme value of $f(x, y, z)$.

g. Multiple Integrals

If $f(x, y)$ is continuous and differentiable at every point within a region 'R' bounded by some curves is given by

$$I = \iint_R f(x, y) dx dy$$

If there is a double integral

$$I = \int_{x=a}^{x=b} \int_{y=\phi(x)}^{y=\psi(x)} f(x, y) dy dx \quad [\text{Let } \psi(x) > \phi(x)]$$

Then I = area of the region bounded by the curves $y = \phi(x)$; $y = \Psi(x)$, $x = a$ and $x = b$ if $f(x, y) = 1$

Value of x – co-ordinate of centroid of the region bounded by $y = \phi(x)$; $y = \psi(x)$; $x = a$, $x = b$

if $f(x, y) = x$

h. Change of Orders of Integration

$$I = \int_{x=a}^{x=b} \int_{y=\phi(x)}^{y=\psi(x)} f(x, y) dy dx \rightarrow I = \int_{x=c}^{x=d} \int_{x=g(y)}^{x=h(y)} f(x, y) dx dy$$

In change of order of Integration, the order of the Integrating variables changes and the limits as well.

